

Part V — Waves, Sound, and Light · Chapter 26

Light Reflection and Refraction

How light behaves at boundaries between media — the foundation of all optical instruments

Press your palm against a glass window and look at your reflection: you can see yourself in the glass and see through it at the same time. The two effects — reflection from the surface and transmission through it — happen simultaneously, with about 4% of the light reflected and 96% transmitted at each glass surface. When a diamond catches the light and scatters a rainbow of colors across a wall, you are seeing two phenomena at once: refraction bending different wavelengths of light by different amounts, and total internal reflection trapping light inside the gem until it can escape only at precisely the right faces. Every optical device ever built — from a magnifying glass to the James Webb Space Telescope — is based on the behavior of light at boundaries. This chapter establishes those principles precisely.

26.1 What Light Is

Light is a form of electromagnetic radiation — a self-sustaining oscillation of electric and magnetic fields that propagates through space at a speed that depends on the medium. In vacuum, light travels at $c = 2.998 \times 10^8$ m/s, universally denoted c and called the speed of light. This is one of the fundamental constants of nature, the same for all observers regardless of their motion — a fact that underpins Einstein’s special theory of relativity.

Visible light is the portion of the electromagnetic spectrum detectable by the human eye, spanning wavelengths from approximately 380 nm (violet) to 700 nm (red). The full electromagnetic spectrum extends from radio waves (wavelengths of kilometers) to gamma rays (wavelengths smaller than an atomic nucleus), all traveling at the same speed c in vacuum and differing only in frequency and wavelength.

Light exhibits a dual nature: under some conditions it behaves as a wave (producing interference and diffraction patterns), and under others as a stream of discrete energy packets called photons. For this chapter, the wave and ray models are sufficient. The photon model is needed for quantum optics and the photoelectric effect, covered in later chapters.

Color	Wavelength (nm)	Frequency (THz)
Violet	380–450	667–789
Blue	450–495	606–667
Green	495–570	526–606
Yellow	570–590	508–526
Orange	590–620	484–508
Red	620–700	428–484

Table 26.1 *Wavelength and frequency ranges for visible colors in vacuum.*

26.2 Light as an Electromagnetic Wave

As established in Chapter 24, electromagnetic waves consist of oscillating electric and magnetic fields perpendicular to each other and to the direction of propagation. Light is transverse: the field oscillations are perpendicular to the direction of travel. This transverse character is what makes polarization possible (Chapter 28).

Because light is an electromagnetic wave, it obeys all the general wave relationships from Chapter 24. The fundamental wave equation $v = f\lambda$ applies: in vacuum, $c = f\lambda$, so wavelength and frequency are linked by $\lambda = c/f$. A violet photon with $\lambda = 400$ nm has frequency $f = c/\lambda = 3 \times 10^8 / 4 \times 10^{-7} = 7.5 \times 10^{14}$ Hz. This is 750 terahertz — the field oscillates 750 trillion times per second.

Unlike sound, light requires no medium. It propagates freely through the vacuum of space. The speed in vacuum is exactly $c = 299\,792\,458$ m/s (the SI definition of the meter is now based on this exact value). When light enters a transparent medium, it slows down. The ratio c/v gives the index of refraction n of that medium (Section 26.9).

Historical Note: The finite speed of light was first estimated by Ole Rømer in 1676 by observing that the orbital period of Io (a moon of Jupiter) appeared shorter when Earth was approaching Jupiter and longer when receding. He inferred this was due to the changing time for light to cross the varying Earth–Jupiter distance, and estimated $c \approx 2.2 \times 10^8$ m/s — within 25% of the correct value, using only a telescope and a pendulum clock.

26.3 Propagation of Light in Vacuum and Media

In a uniform, isotropic medium (one with the same properties in all directions), light travels in straight lines. This is the basis of the ray model of light: we represent a beam of light as a geometric ray — a directed line segment pointing in the direction of propagation. Ray optics (also called geometric optics) is an excellent approximation when the wavelength is much smaller than any obstacles or apertures the light encounters, which is true for most everyday optical situations.

In vacuum, the ray model is exact: light always travels in a straight line at speed c . In a homogeneous medium (uniform composition and density), light also travels in straight lines, but at a reduced speed $v = c/n$, where n is the index of refraction. Speed changes only when the medium changes.

In a non-uniform medium (such as the atmosphere with temperature gradients, or glass with stress patterns), the index of refraction varies from point to point. Rays follow curved paths through such media, continuously bending toward regions of higher n (slower speed). This is the physics behind mirages, the twinkling of stars, and the atmospheric bending that keeps radio signals from escaping into space.

The Rule: Ray model validity: geometric optics applies when $\lambda \ll d$, where d is the size of any aperture or obstacle. For visible light ($\lambda \approx 500$ nm), geometric optics works for objects larger than about 0.1 mm. For smaller features, wave optics (Chapter 28) is required.

26.4 Reflection of Light

Reflection is the redirection of light back into the medium from which it came when it strikes a surface or interface. All surfaces reflect some light; the fraction reflected depends on the optical properties of both media and the angle of incidence.

Specular reflection occurs from smooth, polished surfaces (mirrors, still water, polished metal). Parallel incoming rays reflect as parallel outgoing rays, producing a clear, undistorted image. The surface appears shiny.

Diffuse reflection occurs from rough surfaces (matte paper, walls, skin, unpolished wood). Parallel incoming rays scatter in many directions because each small patch of the surface faces a slightly different direction. No coherent image is formed. The surface appears dull or matte. Most surfaces in everyday life produce diffuse reflection, which is why we can see them from many viewing angles without them appearing as mirrors.

Partial reflection occurs at the interface of two transparent media (air–glass, water–air). Some light is reflected back, and some is transmitted into the second medium. The

fraction reflected depends on the index of refraction difference and the angle of incidence. For normal incidence at an air–glass interface ($n_{\text{glass}} \approx 1.5$), about 4% of the light is reflected and 96% transmitted. This partial reflection is why windows act as both transparent surfaces and weak mirrors.

26.5 The Law of Reflection

The law of reflection governs the direction of specular reflection. It states that the angle of reflection equals the angle of incidence, both measured from the normal to the reflecting surface at the point of incidence.

$$\theta_r = \theta_i$$

Law of Reflection. θ_i = angle of incidence; θ_r = angle of reflection. Both measured from the normal to the surface, not from the surface itself.

The normal is an imaginary line perpendicular to the surface at the point where the ray strikes. It is always measured from the normal, not from the surface. A common mistake is to measure angles from the surface; this gives the complement of the correct angle.

The incident ray, the reflected ray, and the normal at the point of incidence all lie in the same plane, called the plane of incidence. The reflected ray is on the opposite side of the normal from the incident ray, in the same plane. This coplanarity is part of the law of reflection.

Example — Reflection angle A ray strikes a mirror at 35° to the surface (i.e., $90^\circ - 35^\circ = 55^\circ$ from the normal). The reflected ray makes 55° with the normal. The angle between incident and reflected rays is $55^\circ + 55^\circ = 110^\circ$.

Study Tip: Always measure reflection and refraction angles from the normal, never from the surface. If a problem gives the angle from the surface, subtract from 90° to get the angle from the normal before applying any formula.

26.6 Plane Mirrors and Image Formation

A plane mirror is a perfectly flat, specularly reflecting surface. When an object is placed in front of a plane mirror, reflected rays diverge from the mirror in a way that appears to an observer as if they are coming from a point behind the mirror. This apparent source location is the image.

The image formed by a plane mirror has four key properties that follow directly from the law of reflection applied geometrically. First, the image is virtual: the reflected rays

diverge and do not actually converge at the image location; the image exists only where rays appear to originate when extended backward. Second, the image is upright (not inverted). Third, the image is the same size as the object (magnification = 1). Fourth, the image distance behind the mirror equals the object distance in front of it: $d_{\text{image}} = d_{\text{object}}$.

Plane mirrors produce left–right reversal, more precisely described as a front–back reversal (the image’s depth axis is reversed relative to the object’s). When you raise your right hand, the image appears to raise its left hand, because the axis pointing toward the mirror is reversed while the lateral axes are unchanged.

Two plane mirrors at an angle θ to each other produce multiple images. For two mirrors at 90° , three images appear. For mirrors facing each other (parallel), an infinite series of images is produced, each dimmer than the last as the intensity is reduced by repeated partial reflection.

Key Point: A virtual image cannot be projected onto a screen because the light rays do not actually pass through the image location. A real image, formed by converging rays, can be projected (as in a movie projector or camera). The mirror in a bathroom forms a virtual image.

26.7 Refraction of Light

Refraction is the change in direction of a ray of light as it crosses the boundary between two media with different speeds of light. When light moves from a medium in which it travels fast into one where it travels slower, the ray bends toward the normal. When it moves into a faster medium, the ray bends away from the normal.

Refraction occurs because different parts of the wave front reach the boundary at different times, causing the wave front to pivot. The part of the wave that enters the new medium first slows down (or speeds up) while the rest of the front continues at the original speed. This differential speed change rotates the wave front, changing the direction of propagation. (This is the wave explanation of Snell’s Law.)

Everyday examples of refraction: a spoon appears bent in a glass of water (the underwater portion is displaced toward the surface in your perception because rays from it bend as they leave the water); a pool appears shallower than it is (light from the bottom bends away from the normal as it enters air, making the bottom appear higher); the apparent position of the setting Sun is slightly above the horizon even after the Sun has geometrically set (atmospheric refraction bends the rays over the horizon).

Refraction is simultaneously transmission: when light is refracted, it is transmitted into

the second medium. The refracted ray and reflected ray coexist at any real interface; their relative intensities depend on the materials and the angle.

26.8 Why Light Bends Between Media

The physical reason for refraction is the change in wave speed. When a wave enters a medium where it travels more slowly, each part of the wave front that crosses the boundary first is slowed before the rest of the front reaches the boundary. The wave front pivots around the point of contact, changing the direction of propagation.

An analogy: imagine a column of soldiers marching in a line across a field, hitting a muddy region at an angle. The soldiers who reach the mud first slow down while the others continue at full speed. The formation rotates — the direction of the column's advance changes. This is exactly what happens to a wave front entering a slower medium.

The frequency of light does not change at a boundary. The energy of each photon ($E = hf$) is conserved as it crosses the interface; the frequency, set by the source, is a fixed property of the photon. What changes is the speed and therefore the wavelength ($\lambda = v/f$). In a medium with index n , the wavelength is $\lambda_n = \lambda_0/n$, where λ_0 is the vacuum wavelength. The color of light perceived by the eye corresponds to frequency (not wavelength), which is why the color does not change even though the wavelength does when light enters glass.

The Rule: At any boundary: frequency is unchanged (set by source). Speed changes ($v = c/n$). Wavelength changes ($\lambda = v/f = c/(nf) = \lambda_0/n$). Direction changes (if not at normal incidence).

26.9 Index of Refraction

The index of refraction (symbol n) of a medium is defined as the ratio of the speed of light in vacuum to the speed of light in the medium:

$$n = c / v$$

Index of refraction. $c = 3.00 \times 10^8$ m/s; v = speed of light in the medium. Since $v \leq c$ always, $n \geq 1$ always.

The index of refraction is always greater than or equal to 1. Vacuum has $n = 1.000$ exactly. Air at standard conditions has $n = 1.000293 \approx 1.00$ (usually approximated as exactly 1 for most calculations). Water has $n \approx 1.33$, meaning light travels at $343/1.33 \approx 2.26 \times 10^8$ m/s in water. Dense flint glass has n up to about 1.9.

Medium	Index of Refraction (n)	Speed of Light (m/s)
Vacuum	1.000 (exact)	2.998×10^8
Air (STP)	1.000293	2.997×10^8
Water (20°C)	1.333	2.249×10^8
Ice	1.309	2.291×10^8
Crown glass	1.52	1.97×10^8
Dense flint glass	1.89	1.59×10^8
Diamond	2.42	1.24×10^8
Sapphire	1.77	1.69×10^8

Table 26.2 *Index of refraction for common materials at visible wavelengths (values are approximate; n varies slightly with wavelength).*

The index of refraction varies slightly with wavelength — a property called dispersion (Section 26.12). Materials with higher indices refract light more strongly and have more dispersion. Diamond's high index (2.42) and high dispersion are responsible for both its brilliant sparkle and its fire (the rainbow colors it scatters).

26.10 Snell's Law

Snell's Law quantifies the direction of the refracted ray. It was established empirically by Willebrord Snell in 1621 and derived theoretically by Huygens from wave principles. It states:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Snell's Law of Refraction. n_1 = index in the incident medium; θ_1 = angle of incidence (from normal); n_2 = index in the refracted medium; θ_2 = angle of refraction (from normal).

All four quantities are necessary. Snell's Law says that $n \sin\theta$ is conserved across the boundary. If $n_2 > n_1$ (entering a denser medium), then $\sin\theta_2 < \sin\theta_1$, so $\theta_2 < \theta_1$ — the ray bends toward the normal. If $n_2 < n_1$ (entering a less dense medium), the ray bends away from the normal.

Step 1: Identify: n_1 , θ_1 , n_2 . Draw the normal at the point of incidence.

Step 2: Apply: $n_1 \sin\theta_1 = n_2 \sin\theta_2 \rightarrow \sin\theta_2 = (n_1/n_2) \sin\theta_1$

Step 3: Compute: $\theta_2 = \arcsin[(n_1/n_2) \sin\theta_1]$

Step 4: Check: If entering denser medium, $\theta_2 < \theta_1$. If entering less dense medium, $\theta_2 > \theta_1$.

Example — Snell's Law calculation Light in air ($\theta_1 = 40^\circ$) enters water ($n = 1.33$). $\sin\theta_2 = (1.00/1.33) \sin 40^\circ = 0.7519 \times 0.6428 = 0.4833$. $\theta_2 = \arcsin(0.4833) = 28.9^\circ$. The ray bends toward the normal as expected (water is denser: $n_2 > n_1$).

Example — Snell's Law reversed Light in glass ($n = 1.52$, $\theta_1 = 25^\circ$) exits into air. $\sin\theta_2 = (1.52/1.00) \sin 25^\circ = 1.52 \times 0.4226 = 0.6424$. $\theta_2 = \arcsin(0.6424) = 40.0^\circ$. The ray bends away from the normal as expected.

26.11 Total Internal Reflection

When light travels from a denser medium (higher n) into a less dense medium (lower n), the refracted ray bends away from the normal ($\theta_2 > \theta_1$). As the angle of incidence increases, the refracted angle increases more rapidly. At a specific critical angle θ_c , the refracted ray would need to travel at 90° from the normal — along the surface. For any angle of incidence greater than θ_c , no refraction occurs at all: the light is entirely reflected back into the denser medium. This is total internal reflection (TIR).

$$\theta_c = \arcsin(n_2 / n_1) \quad (n_1 > n_2)$$

Critical angle for total internal reflection. Light traveling from medium 1 (index n_1) to medium 2 (index $n_2 < n_1$). TIR occurs for all angles $\theta > \theta_c$.

For glass ($n = 1.52$) to air ($n = 1.00$): $\theta_c = \arcsin(1.00/1.52) = \arcsin(0.658) = 41.1^\circ$. Any ray inside the glass striking the glass–air surface at more than 41.1° from the normal is totally reflected.

For diamond ($n = 2.42$) to air: $\theta_c = \arcsin(1/2.42) = \arcsin(0.413) = 24.4^\circ$. Diamond's very small critical angle means that light entering the gem from almost any direction is trapped inside and undergoes many internal reflections before finally escaping from one of the carefully angled facets. This is why diamonds sparkle so brilliantly: the facets are cut at angles specifically designed to direct internally reflected light upward toward the observer.

Optical fibers exploit TIR to guide light around curves without loss. A glass or plastic fiber with a high-index core and a lower-index cladding confines light within the core by TIR at every surface the ray reaches. As long as the bend radius of the fiber is not too sharp, virtually 100% of the light is guided along the fiber. This enables high-speed internet connections (light pulses carrying data), medical endoscopes (imaging inside the body), and laser delivery for surgery.

Historical Note: John Tyndall demonstrated light guiding by TIR in 1870, directing a beam of light through a curved stream of water and showing that the light followed the curve. This demonstration — light appearing to bend around a corner — was the foundational experiment of fiber optic technology, over a century before fiber optics became commercially practical.

26.12 Dispersion and Prisms

Dispersion is the variation of the speed of light (and therefore the index of refraction) with wavelength within a medium. Most transparent materials have higher indices of refraction for shorter wavelengths (violet) than for longer wavelengths (red). This means violet light bends more than red light when refracted.

When white light (a mixture of all visible wavelengths) passes through a prism, each wavelength refracts by a slightly different angle. After two refractions (entering and exiting the prism), the wavelengths are spread out into a spectrum. Violet light emerges at the largest angle from the original direction; red light at the smallest. The result is the familiar rainbow of colors produced by a glass prism.

Natural rainbows arise from the same physics. Sunlight enters raindrops, refracts at the entry surface, reflects off the back of the drop (TIR or partial reflection), and refracts again on exit. Each wavelength exits at a slightly different angle. Red light exits at about 42° from the anti-solar point (the point directly opposite the Sun from the observer); violet at about 40° . The observer sees a circular arc with red on the outside and violet on the inside.

Dispersion is a practical challenge in optical systems: a simple lens refracts different colors to slightly different focal points, causing colored fringing around images (chromatic aberration). Telescope and camera lens designers use combinations of glasses with different dispersion characteristics (achromatic doublets) to correct this, bringing at least two wavelengths to the same focus.

26.13 Atmospheric Refraction and Mirages

The atmosphere is not uniform: air density (and therefore index of refraction) decreases with altitude. Rays from astronomical objects outside the atmosphere are continuously bent toward Earth as they travel through the increasingly dense lower atmosphere. This atmospheric refraction makes celestial objects appear higher above the horizon than they actually are.

The effect is largest near the horizon, where the air path is longest. The Sun and Moon

appear about 0.5° higher than their actual positions, which equals almost exactly one solar or lunar diameter. This means that when you see the bottom of the Sun touching the horizon, the entire Sun has actually set geometrically — you are seeing it above its true position due to atmospheric refraction.

Mirages are optical illusions caused by atmospheric refraction in non-standard temperature gradients. In a hot desert (or over a hot road), the air near the surface is much hotter and less dense than the air above, creating a gradient in which the index of refraction decreases toward the ground. Rays from the sky travel downward and then curve back upward due to this gradient, creating TIR-like behavior at the hot surface layer. The observer sees an image of the sky below the horizon — which the brain interprets as a pool of water reflecting the sky.

An inferior mirage (the most common type, seen on hot roads) shows an image below the actual object. A superior mirage occurs when cold air lies below warmer air (a temperature inversion): rays curve downward, bringing images of objects over the horizon into view. Mirages of ships and coastlines at extreme distances have been observed in the Arctic and over large cold lakes.

26.14 Common Mistakes in Geometric Optics

- Measuring angles from the surface instead of the normal. The law of reflection and Snell's Law both require angles measured from the normal. If given the angle from the surface, subtract from 90° before using any formula.
- Forgetting that Snell's Law is symmetric. You can apply it in either direction: $n_1 \sin\theta_1 = n_2 \sin\theta_2$ whether light goes from 1 to 2 or from 2 to 1. The medium with the larger n always has the smaller angle.
- Applying TIR to the wrong direction. TIR only occurs when light tries to move from a higher- n medium into a lower- n medium. Light going from air into glass cannot undergo TIR at the air-glass boundary.
- Confusing wavelength and frequency when crossing a boundary. Frequency is unchanged. Wavelength changes ($\lambda_{\text{medium}} = \lambda_0/n$). Color corresponds to frequency (unchanged); wavelength in the medium is shorter than in vacuum.
- Forgetting that n depends on wavelength (dispersion). A single value of n applies only approximately. In precise calculations, n must be specified for the wavelength in question. This is why Snell's Law gives slightly different angles for red and violet light.
- Using $n < 1$ or $v > c$. The index of refraction is always ≥ 1 for any real medium at visible wavelengths. Light slows down, never speeds up, when entering a material medium.

Common Mistake: When applying Snell's Law, always identify which medium is 1 (incident) and which is 2 (refracted) before substituting. A common error is to swap n_1 and n_2 while keeping the correct angles, producing an inverted (and physically wrong) result.

Chapter Summary

Light is an electromagnetic wave that travels at $c = 3.00 \times 10^8$ m/s in vacuum and at $v = c/n$ in a material medium, where n is the index of refraction.

- Visible light spans 380–700 nm in wavelength. Color corresponds to frequency (unchanged at boundaries); wavelength changes with speed: $\lambda = v/f = \lambda_0/n$.
- Specular reflection: $\theta_r = \theta_i$ (both from normal). Diffuse reflection scatters light in all directions from rough surfaces.
- Plane mirrors produce virtual, upright, same-size images at $d_{\text{image}} = d_{\text{object}}$ behind the mirror.
- Refraction: direction change at a speed boundary. Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$. The ray bends toward the normal when entering a denser medium.
- $n = c/v \geq 1$. Higher n means slower speed and stronger bending. Diamond ($n = 2.42$) bends light more than water ($n = 1.33$).
- TIR occurs when light in a denser medium hits a boundary at more than the critical angle: $\theta_c = \arcsin(n_2/n_1)$. Exploited in optical fibers and gem cutting.
- Dispersion: n varies with wavelength, causing different colors to refract at different angles. Responsible for prism spectra and rainbows.
- Atmospheric refraction: non-uniform air bends rays, making objects near the horizon appear higher than their true positions. Mirages arise from extreme temperature-gradient refraction.

Key Terms

electromagnetic radiation: Self-sustaining oscillation of electric and magnetic fields propagating through space; light is a form of electromagnetic radiation.

speed of light (c): The speed of light in vacuum: $c = 2.998 \times 10^8$ m/s; fundamental constant of nature; the maximum speed in the universe.

visible spectrum: The range of electromagnetic wavelengths detectable by the human

eye: approximately 380–700 nm.

ray model (geometric optics): Representation of light as straight directed lines (rays) perpendicular to wave fronts; valid when wavelength $\ll$ obstacle size.

specular reflection: Mirror-like reflection from a smooth surface; parallel incoming rays produce parallel outgoing rays; $\theta_r = \theta_i$.

diffuse reflection: Scattering of light in many directions from a rough surface; no coherent image formed.

normal: A line perpendicular to a surface at the point of incidence; reference line for measuring angles of incidence, reflection, and refraction.

law of reflection: $\theta_r = \theta_i$; the angle of reflection equals the angle of incidence, both measured from the normal.

virtual image: An image from which reflected or refracted rays appear to diverge but through which no light actually passes; cannot be projected onto a screen.

refraction: The change in direction of a wave crossing a boundary between two media with different wave speeds; governed by Snell's Law.

index of refraction (n): $n = c/v$; the ratio of the speed of light in vacuum to its speed in a medium; always ≥ 1 .

Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$; the quantitative relationship between angles and indices at a refracting boundary.

total internal reflection (TIR): Complete reflection of light back into a denser medium when the angle of incidence exceeds the critical angle $\theta_c = \arcsin(n_2/n_1)$.

critical angle (θ_c): The minimum angle of incidence (in the denser medium) for which TIR occurs.

optical fiber: A thin, flexible glass or plastic fiber that guides light along its length by total internal reflection at the core–cladding boundary.

dispersion: The variation of index of refraction with wavelength; causes different colors to refract at different angles and separates white light into a spectrum.

chromatic aberration: The failure of a simple lens to focus different wavelengths to the same point due to dispersion.

mirage: An optical illusion caused by atmospheric refraction in a temperature gradient, producing images of objects in incorrect positions.

Concept Review Questions

Answer in complete sentences. No calculations required unless specified.

1. Why does a swimming pool appear shallower than it actually is? Draw a ray diagram to support your explanation.
2. Explain why the color of light does not change when it crosses a glass surface, even though the wavelength does. What property of the light corresponds to color?
3. A smooth lake surface acts as a mirror in calm weather but not in choppy weather. Explain the physics difference between these two cases.
4. Explain why optical fibers can guide light around curves. What would happen if the fiber were bent too sharply?
5. Why do rainbows always appear as arcs centered on the anti-solar point? Why is red on the outside and violet on the inside?
6. Total internal reflection can only occur from inside the denser medium. Explain why TIR cannot happen when light goes from air into glass.
7. A diamond is cut with many small flat facets at specific angles. Explain using TIR why this maximizes the brilliance of the stone.
8. Explain what a mirage is. Is the mirage an image of water on the road, or an image of the sky? How does the physics of refraction in a temperature gradient produce it?

Reflection Problems

9. A ray of light strikes a flat mirror at 28° from the surface. (a) What is the angle of incidence? (b) What is the angle of reflection? (c) What is the angle between the incident and reflected rays?
10. An object stands 0.40 m in front of a plane mirror. Where is the image? Is it real or virtual? Is it upright or inverted?
11. Two plane mirrors are arranged at 60° to each other. How many images of an object placed between them will appear? (Formula: number of images = $360^\circ/\theta - 1$ for θ dividing 360 evenly.)
12. A ray travels horizontally and strikes a mirror tilted 25° from the vertical. Find the direction of the reflected ray relative to the horizontal.
13. A person 1.70 m tall stands 2.0 m from a plane mirror. What is the minimum mirror length needed so the person can see their full reflection? Does the distance from the mirror matter? Explain.

Refraction Problems

14. Light in air strikes water ($n = 1.33$) at 50° from the normal. Find the angle of refraction in the water.
 15. Light inside glass ($n = 1.52$) hits the glass–air surface at 20° from the normal. Find the angle of refraction in air.
 16. A ray of light enters a flat glass slab ($n = 1.60$) from air at 45° from the normal. (a) Find the angle inside the glass. (b) Find the angle as it exits the opposite parallel face back into air. Explain why the exit angle equals the original entry angle.
 17. What is the speed of light in diamond ($n = 2.42$)?
 18. Light travels from water ($n = 1.33$) into ice ($n = 1.31$). A ray hits the boundary at 30° . Find the refracted angle. Does the ray bend toward or away from the normal?
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Snell's Law Exercises

19. Find the critical angle for the glass–air interface ($n_{\text{glass}} = 1.52$, $n_{\text{air}} = 1.00$).
 20. Find the critical angle at a glass–water interface ($n_{\text{glass}} = 1.52$, $n_{\text{water}} = 1.33$). Is TIR possible at this interface? Under what conditions?
 21. A diamond ring is submerged in water. Find the critical angle for diamond–water TIR ($n_{\text{diamond}} = 2.42$, $n_{\text{water}} = 1.33$). Compare to the diamond–air critical angle (24.4°). What implication does this have for the diamond's appearance underwater?
 22. White light enters a crown glass prism ($n_{\text{red}} = 1.513$, $n_{\text{violet}} = 1.532$) at 30° . Find the refracted angles for red and violet light. What is the angular separation (dispersion angle) between them inside the glass?
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Applied Optics Problems

23. A fish at depth 1.20 m in water looks straight up. What is the radius of the circular patch of the surface through which it can see the entire above-water world? (Hint: use the critical angle for water–air: $\theta_c = \arcsin(1/1.33) = 48.7^\circ$.)
24. An optical fiber has a core with $n = 1.62$ and cladding with $n = 1.50$. Find the critical angle. What is the maximum angle (from the fiber axis) at which a ray can enter the fiber end face and still undergo TIR? (This is the acceptance angle: $\sin(\theta_{\text{accept}}) = \sqrt{(n_{\text{core}}^2 - n_{\text{clad}}^2)}$.)
25. The Sun is actually 0.53° below the geometric horizon when an observer sees it touching the horizon due to atmospheric refraction. How long (in minutes) has the Sun actually set before it appears to touch the horizon? (Earth rotates 360° in $24 \text{ h} = 0.25^\circ$ per minute.)
26. A flat glass window ($n = 1.52$, thickness 4.0 mm) has a ray striking at 40° . Find the perpendicular displacement of the ray as it exits the far side parallel to the original direction. (Use geometry: displacement = thickness $\div \cos\theta_2 \times \sin(\theta_1 - \theta_2)$.)

Chapter 26 Checkpoint

Before moving on to Chapter 27, confirm you can do each of the following.

I can...	See Section
State the speed of light in vacuum and explain what the index of refraction measures	26.1–26.2
Explain when the ray model is valid and when wave optics is needed	26.3
Distinguish specular from diffuse reflection and explain partial reflection	26.4
Apply the law of reflection and measure angles correctly from the normal	26.5
Describe the four image properties of a plane mirror	26.6
Explain refraction qualitatively using the wave-front pivoting argument	26.7–26.8
Calculate the speed of light in a medium given its index of	26.9

refraction	
Apply Snell's Law to find angles of refraction; identify toward/away from normal	26.10
Derive and apply the critical angle formula for TIR	26.11
Explain dispersion and how it produces rainbows and chromatic aberration	26.12
Explain mirages using atmospheric refraction	26.13
Identify and avoid the most common errors in geometric optics problems	26.14

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